

Max Song

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EDUCATION

The University of Texas at Austin | Austin, TX

December 2027

Bachelor of Science in Mechanical Engineering | Design and Manufacturing Track | GPA: 3.8

Relevant Coursework: Mechanics of Solids, Engineering Dynamics, Materials Science, Fluid Mechanics, Computational Methods, Engineering Statics, Thermodynamics

SKILLS

CAD/CAE: SolidWorks (Assemblies, Drawings, BOM), Fusion 360 (CAM), Autodesk Inventor, Blender, Abaqus FEA, ANSYS, MS Office
Analysis: MATLAB, Python (OpenCV, Matplotlib, Pandas, NumPy), FEA (Static, Modal), GD&T, Tolerance Stack Analysis, Structural Analysis

Prototyping: DFM/DFA, Composite Layups (Vacuum Bagging, Resin Infusion), Mechanisms (Gears, Actuators, Bearings, Fasteners)

Manufacturing: CNC Mill, Manual Mill, Manual Lathe, Laser Cutting, SLA and FDM Printing, Soldering, Injection Molding, Table Saw

Materials: Carbon Fiber, Fiberglass, Basalt Fiber, Al 6061-T6, PLA/PETG/TPU/ABS/PA-CF, Elastomers, PMMA

EXPERIENCE

Contoro Robotics - Mechanical Engineering Co-op | Austin, TX

January 2026 - August 2026

- Designed and released electromechanical assembly drawings and full BOMs in SolidWorks across Rev D/E robotic gripper systems (junction boxes, power conduit, motor-driver brackets, camera towers), resolving conveyor-cover clearance and IO-Link mount fit issues to reduce production rework
- Engineered an end-of-line gripper-assembly test stand and test fixture emulating robot hardware, integrating an AL1340 IO-Link master with Festo manifolds, pressure transducers, and pillow-block bearings for pneumatic/vacuum gripper validation
- Built full-scale warehouse unloading robots and commissioned 3 autonomous mobile robots at an Amazon fulfillment center (camera-tower wiring, drive-motor replacement, caster retrofit), retrofitting 4 robots across the D3-to-E2 chassis generation to production spec

Texas Rocket Engineering Lab | Austin, TX

September 2024 - Present

Halcyon Mark 1 - Mechanical Structures Engineer; Composites Manufacturing Engineer

- Led development of 48-inch fiberglass nosecone in SolidWorks, translating DFM requirements into a composite manufacturing process using a 3D-printed ABS vacuum-forming mold; diagnosed and resolved large-format FDM printer mechanical failures to maintain tooling schedule
- Fabricated flight-critical structural components (carbon fiber skirts and couplers, fiberglass raceway mounts) via hand layup and vacuum bagging for a liquid bipropellant spaceshot rocket, achieving 9,600 lbf axial structural validation

Orbital Test Stands - Structures Engineer

- Spearheaded SolidWorks CAD model of a 30-ft hot-fire engine test stand with 30+ components (steel grate flooring, I-beams, brackets, anchors) for Preliminary Design Review (PDR), enabling full structural integration
- Developed Excel-based ACI 318 bolt calculator achieving 150% load capacity with reduced fastener count, meeting GSE safety factors

Texas Inventionworks Makerspace - University of Texas at Austin | Austin, TX

August 2024 - Present

Student Technician - Woodshop Lead and Machine Shop Instructor

- As woodshop lead, directed cross-functional team of 10+ student staff to develop preventive maintenance (PM) schedules, safety protocols, and SOPs for 12+ machines, increasing equipment uptime by over 20%
- Instructed 100+ students in hands-on curriculum covering manual mill, lathe, CNC, and SLA/FDM 3D printing on machine shop team

Architected Intelligent Matter Lab - University of Houston | Houston, TX

June - September 2025

Materials Research Intern

- Developed layered casting method for thin-shell elastomer half-spheres using precision steel ball mold with laser-cut acrylic jig, achieving +/-0.2 mm dimensional accuracy for acoustical metamaterial research
- Improved Abaqus FEA modal simulation accuracy by 10% by calibrating material models using Instron ASTM D412 tensile test data and OpenCV video extensometry; automated the characterization workflow in Python, cutting analysis time 60%

PROJECTS

Custom Macropad Keyboard - Electronics and Product Design Engineer

November 2024

- Designed complete custom assembly and PCB layout, performing component selection, circuit design, and tolerance analysis; manufactured via FDM/SLA 3D printing, hand soldering, and Python firmware with electrical validation testing